



3.6 Core Logic

The `domain_check()` function is the core of the Email Validation System. It validates the domain format by checking for a single dot, non-empty alphanumeric domain and TLD parts, and a valid TLD length. This modular approach makes the validation logic simple, reusable, and easy to maintain.

```
def domain_check(domain):  
    # Must have exactly one dot, and only letters/digits before and after  
    if domain.count('.') != 1:  
        return False  
    local, tld = domain.split('.')  
    if not local or not tld:  
        return False  
    if not local.isalnum() or not tld.isalnum():  
        return False # reject symbols/spaces in domain parts  
    if len(tld) < 2:  
        return False # TLD must be at least 2 characters  
    return True
```

Note: the system currently restricts valid emails to a fixed TLD set and alphanumeric-only local/domain parts, so addresses using hyphens, subdomains, or newer TLDs are rejected; broadening this coverage is planned as a future enhancement.

3.7 Output Screenshot

